MILITARY SPECIFICATION

HOSE, SYNTHETIC RUBBER, WIRE REINFORCED, AND END FITTINGS, REUSABLE, FOR FLEXIBLE HOSE ASSEMBLIES: (FOUR HEAVY SPIRAL WIRE FOR HIGH PRESSURE PHOSPHATE ESTER IN SIZES -20, -24 AND -32)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers synthetic rubber hoses reinforced by four heavy wire reinforcements in sizes -20, -24 and -32 for high pressure phosphate ester service.

1.2 Classification. End fittings for hoses covered by this specification shall be sizes -20, -24 and -32 as applicable and shall be of the following types:

Type AS - Hose sizes -20, -24, and -32 (see figure 1).
Type FS - Hose sizes -20, -24, and -32 (see figure 2).
Type FLS - Hose sizes -20, -24, and -32 (see figure 3).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4720
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SPECIFICATIONS

MILITARY

MIL-H-19457 - Hydraulic Fluid, Fire-Resistant, Non-Neurotoxic.
MIL-H-24135 - Hose, Synthetic Rubber, Wire Reinforced, and End Fittings, Reusable, for Flexible Hose Assemblies, General Specification for.

(Copies of specifications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 582 - Standard Specification for Free-Machining Stainless and Heat-Resisting Steel Bars, Hot-Rolled or Cold Finished. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Hose. Hose furnished under this specification shall conform to the requirements of MIL-H-24135 and as specified herein.

3.2 Material.

3.2.1 Hose. The rubber compounds used shall be resistant to phosphate ester lubricants and shall consist of synthetic rubber or a blend of synthetic rubbers and shall meet the requirements of MIL-H-24135 and as specified herein.

3.2.2 Fittings. Material for fittings shall be as shown on figures 1 through 3.

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3.3 Reinforcement. The reinforcement shall consist of four heavy wire spiral layers applied in such a manner as to meet all the requirements of this specification.

3.4 Dimensions and pressures. Hose dimensions and pressures shall be in accordance with table I.

TABLE I. Dash numbers, performance characteristics, and dimensions.

Hose size (dash no.)	Pressure (lb/in^2)		Hose dimensions (inches)						Min bend radius (inches)
			Id tube		Od cover		Od reinf		
	Working (max)	Burst (min)	min	max	min	max	min	max	
-20	3,000	12,000	1.235	1.297	1.938	2.062	1.781	1.859	18
-24	2,500	10,000	1.485	1.547	2.188	2.312	2.031	2.109	22
-32	2,250	8,000	1.985	2.047	2.688	2.812	2.531	2.609	28

3.5 Fitting types and sizes. Fitting types and sizes shall be as shown on figures 1 through 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-H-24135 and as specified herein.

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4.2 Qualification samples for testing.

4.2.1 Hose. Qualification testing of hose shall consist of hose samples as specified in table II. Samples shall be not less than 15 inches in length and shall be fitted with qualified end fittings.

TABLE II. Numbers of hose samples for qualification testing.

Hose size (dash no.)	Number of hose samples to be provided
-20	4
-32	5

4.2.2 End fittings. Qualification testing of end fittings only shall consist of 18 end fittings of the same type as specified in table III. End fitting samples shall be installed on nine qualified hose samples, in the sizes listed in table II.

TABLE III. Numbers of end fitting samples for qualification testing.

End fitting size (hose dash no.)	Number of end fitting samples to be provided
-20	8
-32	10

4.2.3 Hose and end fittings combined. When a single contractor of both hose and fittings qualifies both hose and fittings, or a hose contractor and a fitting contractor qualifies the hose and fittings in a combined qualification test, the samples shall be provided as specified in table IV.

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TABLE IV. Number of samples for combined qualification testing.

One approved source					
Hose size (dash no.)	No. of hose samples to be provided		End fitting (hose dash no.)	No. of end fittings to be provided	
	Qualifier 2/	Approved 1/ source		Qualifier 2/	Approved 1/ source
-20	2	2	-20	4	4
-32	3	2	-32	5	5

Two or more approved sources							
Hose size (dash no.)	Number of samples to be provided by			End fitting size (dash no.)	Number of end fittings to be provided by		
	Qualifier 2/	Approved 1/ source no. 1	Approved 1/ source no. 2		Qualifier 2/	Approved 1/ source no. 1	Approved 1/ source no. 2
-20	2	1	1	-20	4	2	2
-32	3	1	1	-32	5	3	2

Hose samples shall be not less than 15 inches in length and shall be coupled on one end with the fitting being submitted for qualification and on the other end with previously qualified end fittings.

1/ Approved source - Previously qualified hose and fittings, or hose or fitting.
2/ Qualifier - Hose and fittings, or hose or fitting to be qualified.

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4.2.4 Sequence of tests on sample hose assemblies shall be as specified in table V.

TABLE V. Sequence of qualification tests on sample assemblies.

	Sample numbers								
	Size (dash no.) -20				Size (dash no.) -32				
Test sequence	1	2	3	4	5	6	7	8	9
Proof pressure (see 4.6.1)1/	X	X	X	X	X	X	X	X	X
Stability test (see 4.6.2)1/	X	X	X	X	X	X	X	X	X
Aging test (see 4.6.3)1/		X	X	X		X	X	X	X
Hydraulic fluid circulation (see 4.6.4)1/		X				X			
Impulse test (see 4.6.5)1/				X				X	X
Burst test (see 4.6.6)1/	X				X				
Cold temperature flexing (see 4.6.8)1/			X				X		

1/ See MIL-H-24135.

4.3 The test fluid shall be phosphate ester oil in accordance with MIL-H-19457 for the aging and hydraulic fluid circulation tests.

4.4 The peak pressure for the dynamic impulse test shall be 150 percent of maximum working pressure (see table I).

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 The packaging requirements shall be in accordance with MIL-H-24135.

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6. NOTES

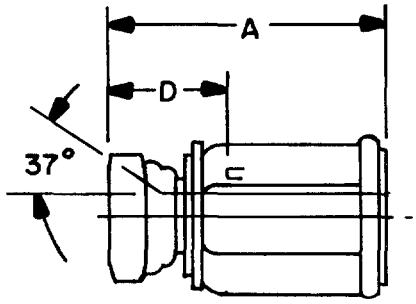
6.1 Intended use. Hose and fittings to this specification are intended for use in shipboard fluid piping systems, utilizing phosphate ester base fluids, where pressures do not exceed the rated working pressure and temperature does not exceed minus 40 to 200°F.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of end fittings for hoses (see 1.2).

Preparing activity:
Navy - SH
(Project 4720-N530-7)

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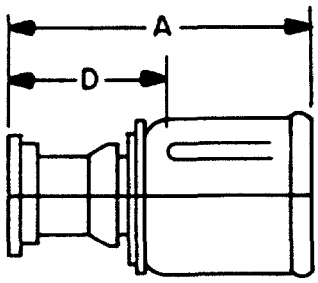
SH 1320266

Thread	Hose size	A (ref)	D (max)
1-5/8-12	-20	4.95	2.33
1-7/8-12	-24	5.60	2.63
2-1/2-12	-32	6.36	2.96

NOTES:
1. Dimensions are in inches.
2. Fitting material shall conform to ASTM A 582, type 303, condition A.

FIGURE 1. Type AS, 37-degree J.I.C. fitting.

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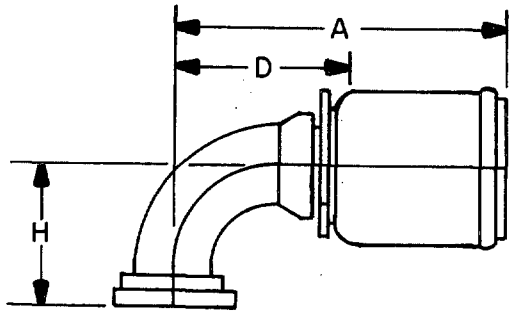
SH 1320267

Flange head size	Hose size	A (max)	D (ref)
-20	-20	5.29	2.67
-24	-24	5.81	2.84
-32	-32	6.30	2.90

NOTES:
1. Dimensions are in inches.
2. Fitting material shall conform to ASTM A 582, type 303, condition A.

FIGURE 2. Type FS.

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SH 1320268

Flange head size	Hose size	A (max)	D (ref)	H (ref)
-20	-20	5.85	3.37	2.62
-24	-24	6.76	3.95	3.12
-32	-32	9.02	5.69	4.38

NOTES:
1. Dimensions are in inches.
2. Fitting material shall conform to ASTM A 582, type 303, condition A.

FIGURE 3. Type FLS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-24135/7(SH)	Hose, Synthetic Rubber, Wire Reinforced, And End Fittings, Reusable, For Flexible Hose Assemblies

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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